Detection and Habitability Potential of Exoplanets Around STAC-1, STAC-2 and STAC-3

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***Abstract*—This paper analyses the luminescence readings of three stars, STAC-1, STAC-2 and STAC-3, to detect transiting exoplanets. STAC-1 and STAC-2 show dips that repeat with a fixed period and constant depth, so each of these stars has signs of hosting a planet. The dips of STAC-3 vary in depth and do not repeat at a fixed period. The datapoints are explained by a star spot rotating with the star. For each planet, the transit depth, orbital period, radius, orbital distance, habitable zone and equilibrium temperature are calculated. STAC-1 b is a Neptune-sized planet very close to its star. STAC-2 b is a super-Earth. STAC-2 b is closer to its habitable zone, cooler and smaller, and is identified as the most promising planet for further study.**

***Index Terms*—transit method, star spots, Kepler's third law, habitable zone**

I. Introduction

The transit method is a common way to detect exoplanets [1]. When a planet passes in front of its star, it blocks a small part of the starlight. The star then appears slightly dimmer for a short time. This event is called a transit.

Dips that repeat at a fixed interval with a constant depth therefore indicate a planet. The interval between dips is the orbital period. Dips that vary in depth or timing usually come from the star itself, for example from star spots.

This paper uses the datapoints given in the problem statement [2]. Each star is checked for transits. The size and orbit of each detected planet are then calculated. Finally, the planets are compared to find the best candidate for habitation.

II. Data and Method

A. Data

Table I lists the properties of the three stars. STAC-1 was measured daily from day 0 to day 20. STAC-2 and STAC-3 were measured from day 0 to day 200, every 20 and 10 days respectively. A brightness of 1.0000 is the normal brightness of the star.

TABLE I
PROPERTIES OF THE STARS

Property	STAC-1	STAC-2	STAC-3
Mass ($M_\odot$)	1.00	0.80	0.50
Radius ($R_\odot$)	1.00	0.80	0.50
Luminosity ($L_\odot$)	1.00	0.40	0.08
Surface temperature (K)	5800	5000	3800

The constants used are $R_\odot = 696{,}000$ km, $R_\oplus = 6{,}371$ km, 1 AU = 149.6×10^6 km and 1 year = 365.25 days [2].

B. Detection of Planet

A star is counted as hosting a planet only if: 1) there are at least three dips, 2) the dips are evenly spaced (within one reading gap), and 3) the dips have the same depth. The transit depth δ is the fraction of light blocked:

$$\delta = (F_{normal} - F_{transit}) / F_{normal} \quad (1)$$

where F_{normal} is the brightness outside transit and $F_{transit}$ is the lowest brightness during transit.

C. Planet Size

The light blocked depends on the size of the planet relative to the star:

$$\delta \approx (R_p / R_s)^2 , \; R_p = R_s \times (\delta)^{1/2} \quad (2)$$

where R_p is the planet radius and R_s is the star radius. Planets are classed as Earth-sized (below 1.5 $R_\oplus$), super-Earth (1.5 to 4 $R_\oplus$), Neptune-sized (4 to 10 $R_\oplus$) or Jupiter-sized (above 10 $R_\oplus$).

D. Orbital Distance

The simple form of Kepler's third law is used, with P in years, a in AU and M in solar masses:

$$P^2 = a^3 / M , \; a = (M \times P^2)^{1/3} \quad (3)$$

E. Habitable Zone and Temperature

The habitable zone is the distance at which a planet receives about the same energy as the Earth receives from the Sun. It is given by [2]:

$$d_{HZ} \approx (L / L_\odot)^{1/2} \text{ AU} \quad (4)$$

The equilibrium temperature is also estimated by taking the ratio of the energy a planet absorbs with the energy it radiates. A planet at 1 AU from the Sun gives about 278 K, so:

$$T_{eq} \approx 278 \times (L / L_\odot)^{1/4} / (a)^{1/2} \text{ K} \quad (5)$$

Here a is in AU. The formula assumes no reflected light and no atmosphere, so it is only an estimate. For comparison, the Earth's actual mean temperature is about 288 K.

III. Results

A. Detection of Exoplanets

Table II summarises the dips found for each star.

TABLE II
DIPS FOUND IN THE BRIGHTNESS READINGS

	STAC-1	STAC-2	STAC-3
Days with a dip	3, 8, 12, 16, 20	20, 100, 180	20, 60, 100, 140, 170
Gap between dips (days)	5, 4, 4, 4	80, 80	40, 40, 40, 30
Brightness at the dips	0.9970 (all)	0.9990 (all)	0.9990 to 0.9993
Same depth every time?	Yes	Yes	No
Fixed period?	Yes	Yes	No
Planet found?	Yes	Yes	No

1) STAC-1: The brightness drops to 0.9970 on days 3, 8, 12, 16 and 20. All five dips have the same depth.

$\delta = (1.0000 - 0.9970) / 1.0000 = 0.0030$ (0.30%)

$P = (20 - 3) / (5 - 1) = 17 / 4 = 4.25$ days

The first gap is 5 days and the rest are 4 days. With daily readings, each transit time is uncertain by up to one day. The average gap of 4.25 days is therefore used.

2) STAC-2: The brightness drops to 0.9990 on days 20, 100 and 180, exactly 80 days apart.

$\delta = (1.0000 - 0.9990) / 1.0000 = 0.0010$ (0.10%)

$P = (180 - 20) / (3 - 1) = 160 / 2 = 80$ days

3) STAC-3: Dips occur on days 20, 60, 100, 140 and 170. These dips are not caused by a planet, for three reasons:

a) *Changing depth.* The dips reach 0.9990, 0.9992, 0.9991, 0.9993 and 0.9990, a change of up to 30%. A planet has a fixed size and blocks the same light on every transit.

b) *Irregular timing.* The first four dips are 40 days apart, but the fifth follows after only 30 days. The 40-day pattern also predicts a dip on day 180, which is absent. A planet's orbit is consistent.

c) *Type of star.* STAC-3 is a cool red dwarf (3800 K, 0.50 $M_\odot$). Such stars have strong magnetic fields and many star spots [3].

A star spot is a cooler, darker patch on the star's surface. It forms where magnetic fields block heat from rising. The spot rotates along with the surface on which it exists. When it faces the observer, the star appears dimmer. When it moves to the far side, the brightness returns to normal. The dip therefore repeats once per rotation [3].

This fits the STAC-3 readings. The dips on days 20, 60, 100 and 140 imply a rotation period of about 40 days. This is typical for red dwarfs, whose measured rotation periods range from under a day to about 70 days [4]. Spots also grow and fade over a few weeks, which explains the changing depth. A new spot forming at another position shifts the timing of the dip. This explains the dip on day 170 and the missing dip on day 180.

Different latitudes of a star also rotate at slightly different speeds, which can change the period further [3].

STAC-3 is therefore excluded from the rest of the analysis.

B. Planet Size

Applying (2) to each planet:

STAC-1 b:

$R_s = 1.00 \times 696{,}000 = 696{,}000$ km

$R_p = 696{,}000 \times (0.0030)^{1/2} = 696{,}000 \times 0.054772$

$R_p = 38{,}121$ km

$R_p / R_\oplus = 38{,}121 / 6{,}371 = 5.98\ R_\oplus$, Neptune-sized

STAC-2 b:

$R_s = 0.80 \times 696{,}000 = 556{,}800$ km

$R_p = 556{,}800 \times (0.0010)^{1/2} = 556{,}800 \times 0.031623$

$R_p = 17{,}608$ km

$R_p / R_\oplus = 17{,}608 / 6{,}371 = 2.76\ R_\oplus$, super-Earth

TABLE III
SIZE OF THE PLANETS

	STAC-1 b	STAC-2 b
Transit depth δ	0.0030	0.0010
Star radius R_s (km)	696,000	556,800
Planet radius R_p (km)	38,121	17,608
Planet radius ($R_\oplus$)	5.98	2.76
Size class	Neptune-sized	Super-Earth

C. Orbital Distance

Each period is converted from days to years and (3) is applied. The distance in km is $a \times 149.6 \times 10^6$.

STAC-1 b:

$P = 4.25 / 365.25 = 0.01164$ years

$a = (1.00 \times 0.01164^2)^{1/3} = (1.355 \times 10^{-4})^{1/3}$

$a = 0.0513$ AU $= 7.67 \times 10^6$ km

STAC-2 b:

$P = 80 / 365.25 = 0.2190$ years

$a = (0.80 \times 0.2190^2)^{1/3} = (0.03837)^{1/3}$

$a = 0.3373$ AU $= 5.05 \times 10^7$ km

The Earth orbits at 1 AU, so a in AU compares each orbit directly with the Earth's (Table IV). STAC-1 b is about 19.5 times closer to its star than the Earth is to the Sun. STAC-2 b is about 3 times closer, slightly inside Mercury's orbit (0.39 AU).

TABLE IV
ORBITS OF THE PLANETS

	STAC-1 b	STAC-2 b
Period P (days)	4.25	80
Period P (years)	0.01164	0.2190
Distance a (AU)	0.0513	0.3373
Distance a (km)	7.67×10^6	5.05×10^7
Compared to Earth	5.1% of Earth's	33.7% of Earth's

D. Habitable Zone and Temperature

Applying (4) to each star:

STAC-1: $d_{HZ} = (1.00)^{1/2} = 1.000$ AU

STAC-2: $d_{HZ} = (0.40)^{1/2} = 0.632$ AU

The ratio a / d_{HZ} shows how close each planet is to its habitable zone. A value of 1 signifies that the planet is in the zone. A value below 1 signifies it is closer to the star, where it is hotter.

STAC-1 b: $0.0513 / 1.000 = 0.051$

STAC-2 b: $0.3373 / 0.632 = 0.534$

Applying (5) for the temperature of each planet:

STAC-1 b:

$(1.00)^{1/4} = 1.000$, $(0.0513)^{1/2} = 0.2265$

$T_{eq} = 278 \times 1.000 / 0.2265 = 1227$ K

STAC-2 b:

$(0.40)^{1/4} = 0.7953$, $(0.3373)^{1/2} = 0.5808$

$T_{eq} = 278 \times 0.7953 / 0.5808 = 381$ K

TABLE V
COMPARISON OF THE TWO PLANETS

	STAC-1 b	**STAC-2 b**
Planet radius ($R_\oplus$)	5.98	2.76
Distance a (AU)	0.0513	0.3373
Habitable zone d_{HZ} (AU)	1.000	0.632
a / d_{HZ}	0.051	0.534
Temperature T_{eq} (K)	1227	381
Rank	2	1

IV. DISCUSSION

A. Most Promising Planet

STAC-2 b is the most promising planet for further habitability studies, for four reasons:

1) *Closer to its habitable zone.* Its a / d_{HZ} is 0.534. For STAC-1 b it is only 0.051, about 20 times too close.

2) *Lower temperature.* Its T_{eq} of 381 K is about 100 K above the Earth's 278 K. STAC-1 b, at 1227 K, is far too hot for liquid water.

3) *Smaller size.* At 2.76 $R_\oplus$ it is less than half the size of STAC-1 b (5.98 $R_\oplus$).

4) *Quiet host star.* STAC-2 is a K-type star. Its dips are clean and regular, with no sign of spots.

B. Limitations

STAC-2 b exists in the inner parts of its habitable zone, with temperatures likely higher than Earth. Being near the habitable zone does not guarantee habitability [2]. Planets larger than about 1.8 $R_\oplus$ often have thick gas envelopes, so STAC-2 b may not be rocky [5]. The 20-day sampling makes the 80-day period an estimate. All formulas used are simplified and give approximate values.

C. Suggested Further Observation

1) Measuring the mass of STAC-2 b would give its density when combined with its radius. The density would show whether the planet is rocky or mostly gas.

2) Observing STAC-2 b with transmission spectroscopy. During a transit, some of the starlight passes through the planet's atmosphere. Gases such as water vapour, carbon dioxide or methane absorb light at certain wavelengths, so the transit depth for each wavelength can be measured. This would denote if the planet has an atmosphere and reveal its composition.

V. CONCLUSION

Planets were detected around STAC-1 and STAC-2 using the transit method. STAC-1 b is Neptune-sized, with a 4.25-day period at 0.0513 AU. STAC-2 b is a super-Earth, with an 80-day period at 0.3373 AU. The STAC-3 dips are inconsistent in depth and timing and denote a star spot. STAC-2 b is closer to its habitable zone, cooler and smaller than STAC-1 b, proving to be the most habitable.

REFERENCES

[1] W. J. Borucki *et al.*, "Kepler planet-detection mission: Introduction and first results," *Science*, vol. 327, no. 5968, pp. 977–980, 2010.

[2] STAC Wars, "PS3 – Exoplanet Hunters: Operation Dim Star," problem statement, 2026.

[3] K. G. Strassmeier, "Starspots," *Astron. Astrophys. Rev.*, vol. 17, no. 3, pp. 251–308, 2009.

[4] A. McQuillan, S. Aigrain, and T. Mazeh, "Measuring the rotation period distribution of field M dwarfs with Kepler," *Mon. Not. R. Astron. Soc.*, vol. 432, no. 2, pp. 1203–1216, 2013.

[5] B. J. Fulton *et al.*, "The California-Kepler Survey. III. A gap in the radius distribution of small planets," *Astron. J.*, vol. 154, no. 3, p. 109, 2017.